



Software Test Engineer

Electives: Automation Testing/ Manual Testing

QP Code: SSC/Q4802

Version: 1.0

NSQF Level: 4

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Qualification Pack

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SSC/Q4802: Software Test Engineer

Brief Job Description

A Software Test Engineer is responsible for designing, developing, and coordinating test plans to ensure software quality and compatibility across diverse environments, including programs, hardware, operating systems, and networks. This role involves performing both scheduled and ad-hoc testing, conducting detailed software compatibility assessments, and identifying potential defects.

Personal Attributes

This role requires strong analytical skills, attention to detail, and the ability to work independently or within teams. The individual must demonstrate logical thinking, effective decision-making, and clear communication while being proactive and result oriented.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N4801: Develop Test Plans for Software Products/ Applications/Modules](#)
2. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Electives(mandatory to select at least one):

Elective 1: Automation Testing

Automation Testing

1. [SSC/N4802: Perform Automated Testing on Software Products/Applications/Modules](#)

Elective 2: Manual Testing

Manual Testing

1. [SSC/N4803: Perform Manual Testing on Software Products/Applications/Modules](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	IT Services

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Occupation	Quality Assurance and Engineering
Country	India
NSQF Level	4
Credits	15
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 2511.0103
Minimum Educational Qualification & Experience	12th grade Pass (Science, Technology, Engineering, and Mathematics (STEM)) with NA of experience OR 10th grade pass with 1.5 years of experience in relevant field. Relevant Experience in Product Development/Software Testing The relevant experience would include work, internship and apprenticeship after completion of relevant educational qualification. OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience (in relevant field. Relevant Experience in Product Development/Software Testing The relevant experience would include work, internship and apprenticeship after completion of relevant educational qualification.)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	1.0
Reference code on NQR	QG-04-IT-03652-2025-V2-NASSCOM
NQR Version	2

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SSC/N4801: Develop Test Plans for Software Products/ Applications/Modules

Description

This National Occupational Standard (NOS) outlines the key responsibilities and competencies required for developing test plans for software products, applications, and modules, ensuring thorough testing and quality assurance across various environments.

Scope

The scope covers the following :

- Analyze Requirements
- Define Test Objectives
- Design Test Cases
- Testing Frameworks and Tools
- Develop the Test Plan Document
- Review and Validate the Test Plan
- Documentation and Reporting
- Manage Test Plan Updates
- Generative AI in Software Testing
- Adhere to Organizational Standards

Elements and Performance Criteria

Analyze Requirements

To be competent, the user/individual on the job must be able to:

- PC1.** review and understand software product/application/module specifications and requirements documents.
- PC2.** identify critical functionalities and performance criteria based on client and project needs.
- PC3.** assess technical feasibility and potential challenges for the testing process.

Define Test Objectives

To be competent, the user/individual on the job must be able to:

- PC4.** establish clear testing objectives, including functional, non-functional, and performance testing requirements.
- PC5.** prioritize test scenarios based on their impact, complexity, and risk assessment.
- PC6.** identify compatibility and compliance standards to be addressed during testing.

Design Test Cases

To be competent, the user/individual on the job must be able to:

- PC7.** create and modify high-level scenarios and detailed test cases, ensuring coverage of positive, negative, and boundary scenarios, and including input conditions, execution steps, and expected outcomes.
- PC8.** utilize reusable and modular testing techniques for scalability and efficiency, accessing test cases, scripts, and tools from the organization's knowledge base.

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Testing Frameworks and Tools

To be competent, the user/individual on the job must be able to:

- PC9.** select and implement appropriate manual and automated testing tools (e.g., Selenium, Appium, JUnit) based on project requirements.
- PC10.** design tests using modern frameworks such as TestNG, Cucumber, or PyTest to ensure scalability and efficiency in testing.

Develop the Test Plan Document

To be competent, the user/individual on the job must be able to:

- PC11.** prepare a comprehensive test plan document detailing scope, objectives, timelines, and resources required.
- PC12.** specify testing tools, environments, and data sets to be used.
- PC13.** outline roles and responsibilities of team members involved in the testing process.

Review and Validate the Test Plan

To be competent, the user/individual on the job must be able to:

- PC14.** review and rework the test plan, test cases, and/or automated scripts with consultation from the supervisor or industry experts.
- PC15.** submit the test plan, test cases, and/or automated scripts for approval by experts.
- PC16.** validate the test plan for completeness, feasibility, and adherence to quality standards.

Documentation and Reporting

To be competent, the user/individual on the job must be able to:

- PC17.** create detailed reports for test execution, results, and defect tracking, providing clear and actionable insights.
- PC18.** use defect tracking tools like Jira, Bugzilla, or Azure DevOps to report and monitor software defects, ensuring timely resolution.

Manage Test Plan Updates

To be competent, the user/individual on the job must be able to:

- PC19.** monitor changes in product/application requirements and update the test plan accordingly.
- PC20.** update the organization's knowledge base with inputs for designing tests and ensure consistency in implementation.
- PC21.** communicate updates to stakeholders and maintain version control for test plan documentation.

Generative AI in Software Testing

To be competent, the user/individual on the job must be able to:

- PC22.** understand the basic concepts and principles of Generative AI, including its potential applications in software testing and quality assurance
- PC23.** identify and explore use cases where Generative AI can enhance software testing, such as test case generation, automated bug detection, and test script optimization
- PC24.** design and optimize effective prompts for Generative AI tools used in test automation, ensuring the accuracy and relevance of generated test cases and scripts.

Adhere to Organizational Standards

To be competent, the user/individual on the job must be able to:

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- PC25.** comply with the organization's testing standards, procedures, and guidelines while designing, executing, and reporting on tests for software products/applications/modules to ensure consistency, quality, and adherence to best practices.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization's policies, procedures, and processes for designing tests for software products/applications/modules, including individual roles and responsibilities in the testing lifecycle.
- KU2.** the application and use of software testing tools, both manual and automated, including when and how to use tools like Selenium, Appium, JUnit, TestNG, and others for specific testing requirements
- KU3.** the different sources of information available for designing tests and how to access these, such as internal knowledge bases, product documentation, user stories, and requirement specifications
- KU4.** the approval process for software products/applications/modules/tests, including who needs to approve test plans, test cases, and scripts and how to submit them for approval.
- KU5.** how to identify issues that may occur with the testing requirements and how to address these, including gaps, ambiguities, or misalignments in the requirements.
- KU6.** the concept and principles of designing tests and ensuring adequate code coverage through appropriate testing strategies, such as boundary value analysis, equivalence partitioning, and error guessing.
- KU7.** how to create suitable tests for applications and how to assess their suitability for automation.
- KU8.** how to access, create, and modify different types of test materials, including high-level scenarios, test cases, automated scripts, test data, and test plans.
- KU9.** the scripting/programming languages to understand test cases and current practices in the industry.
- KU10.** the industry-standard test case design practices and test automation frameworks.
- KU11.** how to track and manage defects through defect tracking tools (e.g., Jira, Bugzilla, Azure DevOps).
- KU12.** the importance of effective defect lifecycle management.
- KU13.** the need for regular maintenance of test scripts, especially in automated testing, to keep them up to date with changing application features and functionalities
- KU14.** how to report and document the results of tests, including creating detailed test execution reports and defect reports.
- KU15.** how to summarize test results and communicate findings with stakeholders.
- KU16.** the best practices for testing in different environments, including compatibility testing across multiple browsers, operating systems, and hardware configurations.
- KU17.** the testing strategies for both functional and non-functional requirements, ensuring that performance, security, and usability aspects are also covered in test plans
- KU18.** the various test methodologies, such as Agile, Waterfall, and DevOps

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to stay informed about the most recent advancements in the field
- GS3.** listen attentively to understand the information/ instructions being shared by the speaker
- GS4.** communicate clearly and politely with co-workers and clients
- GS5.** coordinate with co-workers to achieve work objectives
- GS6.** plan and prioritize tasks to ensure timely completion
- GS7.** identify possible disruptions to work and take appropriate preventive measures
- GS8.** take quick decisions to deal with workplace emergencies/ accidents
- GS9.** evaluate all possible solutions to a problem to select the best one

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Analyze Requirements</i>	3	5	-	2
PC1. review and understand software product/application/module specifications and requirements documents.	1	2	-	1
PC2. identify critical functionalities and performance criteria based on client and project needs.	1	1.5	-	0.5
PC3. assess technical feasibility and potential challenges for the testing process.	1	1.5	-	0.5
<i>Define Test Objectives</i>	3	6	-	3
PC4. establish clear testing objectives, including functional, non-functional, and performance testing requirements.	1	2	-	1
PC5. prioritize test scenarios based on their impact, complexity, and risk assessment.	1	2	-	1
PC6. identify compatibility and compliance standards to be addressed during testing.	1	2	-	1
<i>Design Test Cases</i>	6	10	-	5
PC7. create and modify high-level scenarios and detailed test cases, ensuring coverage of positive, negative, and boundary scenarios, and including input conditions, execution steps, and expected outcomes.	3	5	-	3
PC8. utilize reusable and modular testing techniques for scalability and efficiency, accessing test cases, scripts, and tools from the organization's knowledge base.	3	5	-	2
<i>Testing Frameworks and Tools</i>	2	4	-	2
PC9. select and implement appropriate manual and automated testing tools (e.g., Selenium, Appium, JUnit) based on project requirements.	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. design tests using modern frameworks such as TestNG, Cucumber, or PyTest to ensure scalability and efficiency in testing.	1	2	-	1
<i>Develop the Test Plan Document</i>	3	6	-	1.5
PC11. prepare a comprehensive test plan document detailing scope, objectives, timelines, and resources required.	1	2	-	0.5
PC12. specify testing tools, environments, and data sets to be used.	1	2	-	0.5
PC13. outline roles and responsibilities of team members involved in the testing process.	1	2	-	0.5
<i>Review and Validate the Test Plan</i>	3	5	-	1.5
PC14. review and rework the test plan, test cases, and/or automated scripts with consultation from the supervisor or industry experts.	1	2	-	0.5
PC15. submit the test plan, test cases, and/or automated scripts for approval by experts.	1	1.5	-	0.5
PC16. validate the test plan for completeness, feasibility, and adherence to quality standards.	1	1.5	-	0.5
<i>Documentation and Reporting</i>	2	3	-	1
PC17. create detailed reports for test execution, results, and defect tracking, providing clear and actionable insights.	1	1.5	-	0.5
PC18. use defect tracking tools like Jira, Bugzilla, or Azure DevOps to report and monitor software defects, ensuring timely resolution.	1	1.5	-	0.5
<i>Manage Test Plan Updates</i>	4	6	-	2
PC19. monitor changes in product/application requirements and update the test plan accordingly.	2	2	-	1
PC20. update the organization's knowledge base with inputs for designing tests and ensure consistency in implementation.	1	2	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. communicate updates to stakeholders and maintain version control for test plan documentation.	1	2	-	0.5
<i>Generative AI in Software Testing</i>	3	4	-	1.5
PC22. understand the basic concepts and principles of Generative AI, including its potential applications in software testing and quality assurance	1	1.5	-	0.5
PC23. identify and explore use cases where Generative AI can enhance software testing, such as test case generation, automated bug detection, and test script optimization	1	1.5	-	0.5
PC24. design and optimize effective prompts for Generative AI tools used in test automation, ensuring the accuracy and relevance of generated test cases and scripts.	1	1	-	0.5
<i>Adhere to Organizational Standards</i>	1	1	-	0.5
PC25. comply with the organization's testing standards, procedures, and guidelines while designing, executing, and reporting on tests for software products/applications/modules to ensure consistency, quality, and adherence to best practices.	1	1	-	0.5
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N4801
NOS Name	Develop Test Plans for Software Products/ Applications/Modules
Sector	IT-ITeS
Sub-Sector	
Occupation	Quality Assurance and Engineering
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

SSC/N4802: Perform Automated Testing on Software Products/Applications/Modules

Description

This National Occupational Standard (NOS) outlines the key responsibilities and competencies required for developing test plans for software products, applications, and modules, ensuring thorough testing and quality assurance across various environments.

Scope

The scope covers the following :

- Set Up and Manage Automated Testing Environment
- Develop and Maintain Automated Test Scripts
- Test Preparation and Planning
- Execute and Monitor Automated Tests
- Analyze Test Results
- Integrate Automated Testing into CI/CD Pipeline
- Exploratory and Ad Hoc Testing
- API, Microservices, and Security Testing
- Cloud-Based and Scalable Testing
- Version Control and Collaboration
- Reporting, Documentation, and Compliance

Elements and Performance Criteria

Set Up and Manage Automated Testing Environment

To be competent, the user/individual on the job must be able to:

PC1. set up the test environment, install necessary tools, and verify compatibility with the software under test

PC2. ensure that the environment is optimized and maintained for effective automated testing.

Develop and Maintain Automated Test Scripts

To be competent, the user/individual on the job must be able to:

PC3. write, maintain, and update automated test scripts based on the test plan, ensuring they are reusable and compatible with version control systems

PC4. keep test scripts in sync with software updates and ensure they remain effective for future testing.

Test Preparation and Planning

To be competent, the user/individual on the job must be able to:

PC5. collaborate with superiors to establish the nature of the testing and identify the appropriate test management tool to be used.

PC6. plan test cases and execution strategies according to project requirements.

Execute and Monitor Automated Tests

To be competent, the user/individual on the job must be able to:

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PC7. run automated test scripts in the defined environment and monitor their execution.

PC8. report any issues or discrepancies and ensure the tests are executed as planned.

Analyze Test Results

To be competent, the user/individual on the job must be able to:

PC9. analyze test results to identify discrepancies, log defects, and contribute to defect management discussions.

PC10. ensure test results are documented and follow up on defect resolution.

Integrate Automated Testing into CI/CD Pipeline

To be competent, the user/individual on the job must be able to:

PC11. integrate automated test scripts into the CI/CD pipeline and monitor their execution within the workflow.

PC12. troubleshoot and resolve issues related to automated tests within the CI/CD environment

Exploratory and Ad Hoc Testing

To be competent, the user/individual on the job must be able to:

PC13. perform exploratory testing to identify defects beyond predefined scripts and test cases.

PC14. adapt testing approaches based on unexpected scenarios and findings.

API, Microservices, and Security Testing

To be competent, the user/individual on the job must be able to:

PC15. design automated test cases for APIs and microservices, validating their functionality and integration.

PC16. design basic security test cases, identify vulnerabilities, and contribute to penetration testing efforts for the application.

Cloud-Based and Scalable Testing

To be competent, the user/individual on the job must be able to:

PC17. utilize cloud-based platforms for testing applications and perform scalability and performance testing in cloud environments.

Version Control and Collaboration

To be competent, the user/individual on the job must be able to:

PC18. manage automated test scripts using version control systems and collaborate with team members on script updates.

PC19. ensure proper version control practices are followed to maintain script integrity.

Reporting, Documentation, and Compliance

To be competent, the user/individual on the job must be able to:

PC20. document test execution results, defects, and generate clear, concise test reports for stakeholders.

PC21. adhere to testing standards, best practices, and comply with organizational testing procedures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the various test management tools and their application in automated testing

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- KU2.** the different types of automated testing tools (e.g., Selenium, Appium, JUnit, TestNG, etc.).
- KU3.** the test script writing, debugging, and maintenance techniques.
- KU4.** the software requirements and test plan creation.
- KU5.** the Continuous Integration/Continuous Deployment (CI/CD) pipelines and integrating automated tests into them
- KU6.** the defect tracking and test result reporting tools (e.g., Jira, Bugzilla, Azure DevOps).
- KU7.** the best practices for creating reusable test scripts.
- KU8.** the cloud-based testing tools and platforms
- KU9.** the API and microservices testing, including the design and execution of automated tests.
- KU10.** the basic security testing concepts and vulnerability assessment techniques.
- KU11.** the exploratory and ad hoc testing methodologies.
- KU12.** the version control systems, particularly Git, and their use in managing test scripts.
- KU13.** the awareness of the latest trends and tools in automated testing for modern software development.
- KU14.** the specific requirements for maintaining the testing environment and ensuring compatibility with the software being tested.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to stay informed about the most recent advancements in the field.
- GS3.** listen attentively to understand the information/ instructions being shared by the speaker
- GS4.** communicate clearly and politely with co-workers and clients
- GS5.** coordinate with co-workers to achieve work objectives
- GS6.** plan and prioritize tasks to ensure timely completion
- GS7.** identify possible disruptions to work and take appropriate preventive measures
- GS8.** take quick decisions to deal with workplace emergencies/ accidents
- GS9.** evaluate all possible solutions to a problem to select the best one

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Set Up and Manage Automated Testing Environment</i>	2	4	-	1
PC1. set up the test environment, install necessary tools, and verify compatibility with the software under test	1	2	-	0.5
PC2. ensure that the environment is optimized and maintained for effective automated testing.	1	2	-	0.5
<i>Develop and Maintain Automated Test Scripts</i>	4	4	-	2
PC3. write, maintain, and update automated test scripts based on the test plan, ensuring they are reusable and compatible with version control systems	2	2	-	1
PC4. keep test scripts in sync with software updates and ensure they remain effective for future testing.	2	2	-	1
<i>Test Preparation and Planning</i>	2	4	-	2
PC5. collaborate with superiors to establish the nature of the testing and identify the appropriate test management tool to be used.	1	2	-	1
PC6. plan test cases and execution strategies according to project requirements.	1	2	-	1
<i>Execute and Monitor Automated Tests</i>	3	4	-	2
PC7. run automated test scripts in the defined environment and monitor their execution.	1.5	2	-	1
PC8. report any issues or discrepancies and ensure the tests are executed as planned.	1.5	2	-	1
<i>Analyze Test Results</i>	4	4	-	2
PC9. analyze test results to identify discrepancies, log defects, and contribute to defect management discussions.	2	2	-	1
PC10. ensure test results are documented and follow up on defect resolution.	2	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Integrate Automated Testing into CI/CD Pipeline</i>	3	6	-	2
PC11. integrate automated test scripts into the CI/CD pipeline and monitor their execution within the workflow.	1.5	3	-	1
PC12. troubleshoot and resolve issues related to automated tests within the CI/CD environment	1.5	3	-	1
<i>Exploratory and Ad Hoc Testing</i>	3	4	-	2
PC13. perform exploratory testing to identify defects beyond predefined scripts and test cases.	1	2	-	1
PC14. adapt testing approaches based on unexpected scenarios and findings.	2	2	-	1
<i>API, Microservices, and Security Testing</i>	2	6	-	2
PC15. design automated test cases for APIs and microservices, validating their functionality and integration.	1	3	-	1
PC16. design basic security test cases, identify vulnerabilities, and contribute to penetration testing efforts for the application.	1	3	-	1
<i>Cloud-Based and Scalable Testing</i>	2	4	-	1
PC17. utilize cloud-based platforms for testing applications and perform scalability and performance testing in cloud environments.	2	4	-	1
<i>Version Control and Collaboration</i>	2	6	-	2
PC18. manage automated test scripts using version control systems and collaborate with team members on script updates.	1	3	-	1
PC19. ensure proper version control practices are followed to maintain script integrity.	1	3	-	1
<i>Reporting, Documentation, and Compliance</i>	3	4	-	2
PC20. document test execution results, defects, and generate clear, concise test reports for stakeholders.	2	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. adhere to testing standards, best practices, and comply with organizational testing procedures	1	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N4802
NOS Name	Perform Automated Testing on Software Products/Applications/Modules
Sector	IT-ITeS
Sub-Sector	
Occupation	Quality Assurance and Engineering
NSQF Level	4
Credits	5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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SSC/N4803: Perform Manual Testing on Software Products/Applications/Modules

Description

This unit outlines the competencies required to perform manual testing of software products, applications, or modules. It includes preparing for testing, executing test cases, analyzing results, and reporting findings, with a focus on ensuring software quality, compatibility, and accessibility.

Scope

The scope covers the following :

- Prepare for Manual Testing
- Execute Manual Test Cases
- Report Test Results and Defects
- Collaborate with Automation Teams
- Ensure Continuous Improvement of Manual Testing Processes

Elements and Performance Criteria

Prepare for Manual Testing

To be competent, the user/individual on the job must be able to:

- PC1.** identify testing requirements, including functionality, compatibility, and accessibility standards.
- PC2.** verify that the latest and correct versions of test cases and related documents are available for execution.
- PC3.** ensure the availability and readiness of test environments, data, and tools.
- PC4.** identify the range of browsers, operating systems, and devices for cross-browser and cross-platform testing.

Execute Manual Test Cases

To be competent, the user/individual on the job must be able to:

- PC5.** execute predefined test cases accurately and within the required timelines.
- PC6.** validate software conformance to usability and accessibility guidelines (e.g., WCAG).
- PC7.** perform exploratory and dynamic testing to identify defects in unscripted scenarios.
- PC8.** execute manual test scripts and take corrective action in case of failures or deviations.
- PC9.** conduct cross-browser testing to verify compatibility with different browsers and versions.
- PC10.** carry out cross-platform testing to validate functionality across operating systems and devices

Report Test Results and Defects

To be competent, the user/individual on the job must be able to:

- PC11.** log detailed test progress, results, and defects in the agreed format and test management tool
- PC12.** communicate critical defects and risks to supervisors and stakeholders promptly

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PC13. document issues with supporting evidence (e.g., screenshots, logs) for efficient resolution.

PC14. analyze defects to determine their root causes and provide detailed feedback to developers

Collaborate with Automation Teams

To be competent, the user/individual on the job must be able to:

PC15. identify scenarios where automation testing might improve efficiency and reduce errors.

PC16. work closely with automation testers to ensure the seamless integration of manual and automated testing processes.

PC17. provide inputs for designing test scripts that cover edge cases or complex workflows identified during manual testing.

Ensure Continuous Improvement of Manual Testing Processes

To be competent, the user/individual on the job must be able to:

PC18. obtain guidance from supervisors or experts for complex or unclear scenarios.

PC19. update and improve test cases based on testing outcomes and feedback

PC20. stay updated on the latest manual testing techniques, tools, and best practices.

PC21. ensure adherence to the organization's policies, procedures, and quality standards

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the organization's policies, procedures, and guidelines for manual testing and the role in applying these.

KU2. the latest and correct versions of test cases and test scripts used for manual testing.

KU3. how to execute manual test scripts correctly and take necessary actions in case of failures

KU4. the awareness of usability and accessibility standards, such as WCAG, and how to verify conformance during testing.

KU5. the cross-browser and cross-platform testing principles and techniques to ensure compatibility across browsers, operating systems, and devices.

KU6. how to log test progress, results, and defects using the agreed test management tool

KU7. how to analyze test results to identify defects and understand their root causes

KU8. the dynamic testing methods to identify defects in unscripted scenarios.

KU9. how to collaborate with automation testers to identify scenarios where automation can improve efficiency.

KU10. how to provide feedback and suggestions for improving manual testing processes

KU11. the awareness of the tools, techniques, and best practices used in manual testing to ensure continuous improvement

KU12. how to communicate defects, issues, and risks to relevant stakeholders effectively

KU13. how to stay updated on the latest trends and developments in manual testing and incorporate them into testing practices

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to stay informed about the most recent advancements in the field.
- GS3.** listen attentively to understand the information/ instructions being shared by the speaker
- GS4.** communicate clearly and politely with co-workers and clients
- GS5.** coordinate with co-workers to achieve work objectives
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- GS7.** identify possible disruptions to work and take appropriate preventive measures
- GS8.** take quick decisions to deal with workplace emergencies/ accidents
- GS9.** evaluate all possible solutions to a problem to select the best one

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for Manual Testing</i>	6	10	-	4
PC1. identify testing requirements, including functionality, compatibility, and accessibility standards.	1.5	2.5	-	1
PC2. verify that the latest and correct versions of test cases and related documents are available for execution.	1.5	2.5	-	1
PC3. ensure the availability and readiness of test environments, data, and tools.	1.5	2.5	-	1
PC4. identify the range of browsers, operating systems, and devices for cross-browser and cross-platform testing.	1.5	2.5	-	1
<i>Execute Manual Test Cases</i>	9	15	-	6
PC5. execute predefined test cases accurately and within the required timelines.	1.5	2.5	-	1
PC6. validate software conformance to usability and accessibility guidelines (e.g., WCAG).	1.5	2.5	-	1
PC7. perform exploratory and dynamic testing to identify defects in unscripted scenarios.	1.5	2.5	-	1
PC8. execute manual test scripts and take corrective action in case of failures or deviations.	1.5	2.5	-	1
PC9. conduct cross-browser testing to verify compatibility with different browsers and versions.	1.5	2.5	-	1
PC10. carry out cross-platform testing to validate functionality across operating systems and devices	1.5	2.5	-	1
<i>Report Test Results and Defects</i>	6	10	-	4
PC11. log detailed test progress, results, and defects in the agreed format and test management tool	1.5	2.5	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. communicate critical defects and risks to supervisors and stakeholders promptly	1.5	2.5	-	1
PC13. document issues with supporting evidence (e.g., screenshots, logs) for efficient resolution.	1.5	2.5	-	1
PC14. analyze defects to determine their root causes and provide detailed feedback to developers	1.5	2.5	-	1
<i>Collaborate with Automation Teams</i>	4	7	-	3
PC15. identify scenarios where automation testing might improve efficiency and reduce errors.	1.5	2.5	-	1
PC16. work closely with automation testers to ensure the seamless integration of manual and automated testing processes.	1.5	2.5	-	1
PC17. provide inputs for designing test scripts that cover edge cases or complex workflows identified during manual testing.	1	2	-	1
<i>Ensure Continuous Improvement of Manual Testing Processes</i>	5	8	-	3
PC18. obtain guidance from supervisors or experts for complex or unclear scenarios.	1.5	2	-	1
PC19. update and improve test cases based on testing outcomes and feedback	1.5	2	-	1
PC20. stay updated on the latest manual testing techniques, tools, and best practices.	1	2	-	0.5
PC21. ensure adherence to the organization's policies, procedures, and quality standards	1	2	-	0.5
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N4803
NOS Name	Perform Manual Testing on Software Products/Applications/Modules
Sector	IT-ITeS
Sub-Sector	
Occupation	Quality Assurance and Engineering
NSQF Level	4
Credits	5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment shall be conducted through an online proctored format, incorporating scenario-based multiple-choice questions designed to effectively evaluate practical understanding and real-world application of concepts. Additionally, it will include a viva-voce and hands-on practical evaluation to comprehensively assess the individual's proficiency in specific learning outcomes.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N4801.Develop Test Plans for Software Products/ Applications/Modules	30	50	0	20	100	70
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	50	80	-	20	150	80

Elective: 1 Automation Testing

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N4802.Perform Automated Testing on Software Products/Applications/Modules	30	50	0	20	100	20
Total	30	50	-	20	100	20

Elective: 2 Manual Testing

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N4803.Perform Manual Testing on Software Products/Applications/Modules	30	50	0	20	100	20
Total	30	50	-	20	100	20

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.